MIL-P-47075(AR)
10 May 1974
SUPERSEDING
MIS-13982
2 April 1969

MILITARY SPECIFICATION

PLASTIC, SHEET, POLYOLEFIN, SPUNBONDED

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one style of spunbonded polyolefin sheet.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament, Research and Development Center, ATTN: SMCAR-ESC-AS, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on data of invitation for bids shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

T-404	Tensile Breaking Strength of Paper and Paperboard
T-410	Basis Weight of Paper and Paperboard
T-411	Thickness of Paper and Paper Products

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York, 10017.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet the requirements of this specification.

3.2 Material

3.2.1 Composition. The sheet shall be manufactured from 100 percent high density linear polyethylene.

3.2.2 Construction. The sheet web-structure shall be a random distribution of very fine continuous fibers (0.02 to 0.04 mils in their smallest dimension) formed without the use of binders, sizes or fillers.

3.2.3 Thickness. The sheet thickness shall be 0.007 inch nominal.

3.3 Performance characteristic. The finished sheet shall conform to the requirements specified in Table I.

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3.4 Bondability. The sheet shall be bondable to itself by use of commercially common heat or electronic sealing devices and methods. The strength of such bonds in either the machine or transverse direction of the sheet shall be not less than 65 percent of the tensile strength of the sheet.

3.5 Workmanship. The workmanship shall be such as to assure a product which is uniform and in accordance with this specification. The sheet shall be clean, well finished, and free from dirt, foreign matter, oil, rough or sharp edges, cracks, creases, blisters, bubbles, pinples, pits, tears, cuts or holes (other than the perforation pattern), which might impair its usefulness for the purpose intended.

TABLE 1

PERFORMANCE CHARACTERISTICS

PROPERTY	TEST PARAGRAPH	REQUIREMENT
Strip Tensile Strength	4.5.3.2	
Machine Direction		6.5 lb/in (min.)
Transverse Direction		10.5 lb/in (min.)
Elongation	4.5.3.2	
Transverse Direction		21 to 31 percent
Basic Weight	4.5.3.4	1.4 to 1.8 oz/sq. yd.
Frazier Porosity	4.5.3.5	10 to 54 cu. ft/sq.ft. (min.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Preproduction inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of 4 square feet of sheet, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. Subsequent units will not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordancae with the correspondence test paragraphs.

TABLE II

PREPRODUCTION INSPECTION

CHARACTERISTICS	REQUIREMENT SOURCE	TEST PARAGRAPH
Strip Tensile	3.3	4.5.3.2
Elongation	3.3	4.5.3.2
Basic Weight	3.3	4.5.3.4
Frazier Porosity	3.3	4.5.3.5
Bondability	3.4	4.5.3.3

4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted on a random sample form each inspection lot or less, in accordance with MIL-STD-105 at an acceptable quality level (AWL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in ne continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1 to determine compliance with the characteristics specified in table III shall be conducted in accordance with the corresponding test and inspection paragraphs.

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TABLE III

ACCEPTANCE INSPECTION		
CHARACTERISTICS	REQUIREMENT SOURCE	TEST PARAGRAPH
Composition	3.2.1	4.5.3.1
Construction	3.2.2	4.5.3.1
Thickness	3.2.3	4.5.3.6
Workmanship	3.5	4.7
Packaging and Packing	5.1	4.6
Marking	5.2	4.6

4.4 Inspection equipment. The inspection equipment for conducting examinations and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedure.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shsll be used as a basis to establish performance requirements:

a. Temperature — Room ambient (16 to 32 degrees Celsius (C)) (60 to 90 degrees Fahrenheit (F)).

b. Altitude — Facility ground.

c. Humidity — Facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Test sequence within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Tests.

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4.5.3.1 Composition and construction. The composition and construction of the material shall be certified by the supplier to be in conformance to 3.2.1 and 3.2.2.

4.5.3.2 Strip tensile strength and elongation. Strip tensile strength and elongation shall be determined in accordance with T-404 and shall conform to 3.3.

4.5.3.3 Bondability. The bonding strength shall be determined in accordance with T-404 and shall conform to 3.4.

4.5.3.4 Basis weight. The basis weight shall be determined in accordance with T-410 and shall conform to 3.3.

4.5.3.5 Frazier porosity. Frazier porosity shall be determined in accordance with FED-STD-191, Method 5450, and shall conform to 3.3.

4.5.3.6 Thickness. Thickness shall be determined in accordance with T-411 and shall conform to 3.2.3.

4.6 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing and marking of material conforms to this specification.

4.7 Workmanship. The material shall be visually examined to assure conformance to the workmanship requirements specified in 3.5.

5. PACKAGING

5.1 Packaging and packing. The material shall be furnished in rolls, as specified by the procuring agency, from the range of length, width, core diameter, allowable tolerances and number of splices normally avialable from the supplier (see 6.2). The material shall be packaged and packed in accordance with accepted industry practice and in such a manner as to preclude damage or contamination in handling, transportation or storage from the source to the first receiving activity.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot or batch number.

d. Date of manufacture.

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e. Number of this specification.

6. NOTES

6.1 Intended use. The material conforming to this specification is intended for use as the corridor in the construction of a moisture-barrier fly-through diaphragm in a missile container for protection against corrosion and deterioration prior to launch.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1) and, if so, pertinent details (see 4.2).

c. Quantity and size (see 5.1).

d. Requirement for certification of composition and construction (see 4.5.3.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13982, dated 2 April 1969.

Custodian:
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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-47075	PLASTIC, SHEET, POLYOLEFIN, SPUNBONDED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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